

APPLICANT BIOGRAPHICAL SKETCH

NAME:	Eshun-Wilson, Lisa
eRA COMMONS USERNAME:	lisaeshunwilson
POSITION TITLE:	Principal Investigator

EDUCATION/TRAINING

INSTITUTION AND LOCATION	DEGREE (if applicable)	Completion Date MM/YYYY	FIELD OF STUDY
Grinnell College	B.A.	05/2014	Biological Chemistry
The National Institutes of Health	Postbaccalaureate	08/2015	Membrane Biology
University of California, Berkeley	Ph.D.	12/2019	Molecular Cell Biology
The Scripps Research Institute	Postdoctoral	04/2024	Integrative Computational and Structural Biology
Profluent Bio, Inc.	Scientist II	08/2025	AI-first protein design
The Genoma Institute	Principal Investigator	Present	Gene Editing

A. Personal Statement

For my doctoral thesis, I chose to tackle a long-standing controversy: whether the relationship between the chemical modification α K40 acetylation and MT stability was causative or correlative. Interestingly, α K40 acetylation is the only post-translational modification (PTM) to occur on the inside of MTs and mark long-lived, stable MT populations, including those that compose the cytoskeleton of a cancerous cell. Moreover, the misregulation of α K40 acetylation is linked to axonal transport defects associated with Huntington's disease, Charcot-Marie-Tooth disease, amyotrophic lateral sclerosis, Parkinson's disease, and the growth of "microtentacles" that promote metastatic breast cancer. To elucidate the potential causative effects of this modification, I applied a reductionist approach to tease out the direct effects of this modification on MT structure by using cryo-electron microscopy (cryo-EM). By visualizing purely acetylated and deacetylated MTs, I showed that acetylation changes the conformational ensemble of the α K40 loop in α -tubulin, or in other words it changes the rhythm of the modification site of this PTM and may serve as an evolutionarily conserved 'electrostatic switch' to regulate MT stability. This work was launched in close collaboration with Drs. James Fraser at UCSF and Max Bonomi at the University of Cambridge.

Similar to PTMs, microtubule associated proteins (MAPs), and therapeutic agents can affect MT structure and stability as well. To build on my passion for intracellular transport sparked by my time at the NIH with Dr. Julie Donaldson, I investigated the mechanism by which microtubule-associated protein 7 (MAP7) modulates kinesin-1 motility. Using cryo-electron microscopy and single-molecule imaging, I showed that the microtubule-binding domain of MAP7 binds as an extended α -helix along the protofilament ridge, partially overlapping with the kinesin-1 binding site and directly inhibiting motor motility. Unexpectedly, MAP7 simultaneously promotes transport by tethering kinesin-1 to the microtubule lattice via its projection domain, preventing motor dissociation and facilitating rebinding to adjacent sites. Together, this work revealed a biphasic, concentration-dependent regulatory mechanism in which MAP7 both inhibits and enhances kinesin-1 transport through competitive and cooperative interactions on the microtubule surface.

Taxol, a major breast cancer chemotherapy agent, can block the cell cycle in its G1 or M phases by stabilizing MTs and limiting MT critical dynamics. Surprisingly, lankacidins (LCs) were shown to have both in vivo antitumor activity in multiple cancer cell lines and antimicrobial activity against Gram-positive pathogens. We observed that the effects of LCs were minimal and focused on uncovering the structural basis of their antimicrobial activity and resistance. I resolved a 2.8 Å structure of the LC-ribosome complex. I discovered that LC forms an elaborate hydrophobic network within the peptidyl transferase center (PTC) of the exit (E) site of the ribosome which is essential for its inhibitory effect on translation and common for this class of macrolides, consistent with previous research. Moreover, we show that the ring closure is important for the inhibitory effect of LC on harmful bacteria. Previous evidence supports that when the macrocyclic ring is hydrogenated its ring conformation changes and its inhibitory effects are reduced. Thus, bacteria may resist LC over time by manipulating the identity of the nucleotides that compose the hydrophobic network, so as not to lethally affect ribosome function, but weaken LC binding to the PTC site. This work was done in collaboration with Drs. James Fraser and Ian Seiple at UCSF.

She then completed a postdoctoral fellowship at the Scripps Research Institute in La Jolla, CA, where she resolved the first cryo-EM structure of Hepatitis C virus (HCV) glycoprotein, resulting in her second first author publication in Science Magazine. She was recruited to Profluent Bio, Inc., an AI-first protein design company, where she worked at the intersection of machine learning, high-throughput sequencing, and AI-led protein design, deepening her understanding of both the power and biosecurity implications of generative biological systems. Now, through Genoma, she is building new scientific, organizational, and ethical frameworks for genome engineering research, emphasizing high-risk tool development, interdisciplinary collaboration, and ethical deployment.

Selected Publications (from this biosketch)

1. Eshun-Wilson L, Zhang R, Portran D, Toso D, Nachury M, Bonomi M, Fraser JS, Nogales E. *Structural insights into the effects of α -tubulin acetylation on microtubule structure and properties*. Proceedings of the National Academy of Sciences (2019) 116 (21) 10366-10371.
2. **Eshun-Wilson L**, Zhang R, Portran D, Toso D, Nachury M, Bonomi M, Fraser JS, Nogales E. *Structural insights into the effects of α -tubulin acetylation on microtubule structure and properties*. **Proceedings of the National Academy of Sciences** (2019) 116(21):10366–10371.
3. **Eshun-Wilson L**, Ferro LS, Fang Q, Fernandes J, Jack A, Farrell DP, Golcuk M, Huijben T, Costa K, Gur M, DiMaio F, Nogales E, Yildiz A. *Structural and functional insight into regulation of kinesin-1 by microtubule-associated protein MAP7*. **Science** (2022) 375(6578):326–331.
4. **Eshun-Wilson L[#]**, Torrents de la Peña A[#], Sliepen K[#], Newby ML, Allen JD, Zon I, Koekkoek S, Chumbe A, Crispin M, Schinkel J, Lander GC, Sanders RW, Ward AB. *Structure of the hepatitis C virus E1E2 glycoprotein complex*. **Science** (2022) 378(6617):263–269.

B. Positions and Honors

Positions and Employment

2014–2015 Intramural Research Training Award Postbaccalaureate Fellow, Membrane Biology Section, National Heart, Lung and Blood Institute, Bethesda, MD

2020–2024 Postdoctoral Fellow, Department of Integrative Computational and Structural Biology, Scripps Research Institute, La Jolla, CA

2024–2025 Scientist II, Profluent Bio, Inc., Berkeley, CA

2024–Present Principal Investigator, Genoma, Berkeley, CA

Other Experience and Professional Memberships

2020–Present Member, Biophysical Society

2014–Present Member, American Society of Cell Biology

2014–Present Member, American Society of Biochemistry and Molecular Biology

Honors

2020 Cris Alvaro PhD Commencement Prize

2019 Carl Storm Underrepresented Minority Fellowship (CSURM), Gordon Research Conference

2019 Summer Conference Travel Grant, UC Berkeley Graduate Division

2019 Chancellor’s Award for Public Service Nominee

2019 Dean of Students Outstanding Leadership Award Nominee

2019 RISE Award Winner, Gender Resource Center

2018 MCB Equity & Inclusion Award

2016 National Science Foundation Graduate Fellowship

2016 Ford Foundation Predoctoral Fellowship, National Academy of Sciences

2014 Travel Award, King Abdullah University of Science and Technology

2014 John. Y. Young Service Memorial Scholarship

2014 Grinnell Ladies Education Society Scholarship

2013 Howard Hughes Medical Institute Summer Fellowship, CalTech

2013 POSSE Foundation Summer Leadership Award

2012 Amgen Scholars Summer Fellowship, UCSF

2010 Fulfillment Fund Higher Education Scholarship

C. Contribution to Science

1. My first publication focused on the role post-translational modifications, enzymes, and therapeutic agents have on microtubule structure, dynamics and regulations, due the important role of these cytoskeletal polymers in cancer, neurogenerative disease and aneuploidy. I was interested in how unique modulators could alter the conformation and plasticity of the microtubule lattice. For instance, the structure and mechanics of microtubules are not only dependent on the modification state of each tubulin, but also on tubulin isotypes, interacting drugs, or MT-binding partners, all of which can cause changes in tubulin structure and subunit packing within the MT and affect local mechanical strain and other physical properties of the lattice. I started to appreciate the MT as an allosteric macromolecular machine that interprets multifaceted inputs and reacts by transforming its rigidity and mechanical resistance. This appreciation launched an exploratory project into the complex world of chemical modifications, specifically on acetylation, the main post-translational modification to occur on the inside of the microtubule, and lankacidins, a unique class of antibiotics. I served as the lead author in all of these studies.

a. Eshun-Wilson L, Zhang R, Portran D, Toso D, Nachury M, Bonomi M, Fraser JS, Nogales E. *Structural insights into the effects of α -tubulin acetylation on microtubule structure and properties*. Proceedings of the National Academy of Sciences (2019) 116 (21) 10366-10371.

b. Eshun-Wilson L., Pellegrino, J. Ward, F., Toso, D., Brilot, A., Nogales, E., Seiple, I., Fraser, J. *CryoEM and synthetic studies reveal that ring closure of lankacidin, and related metabolites mediates antibiotic activity*. Manuscript in preparation.